POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2019
Programme: BE | Full Marks: 100
Course: Artificial Intelligence | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

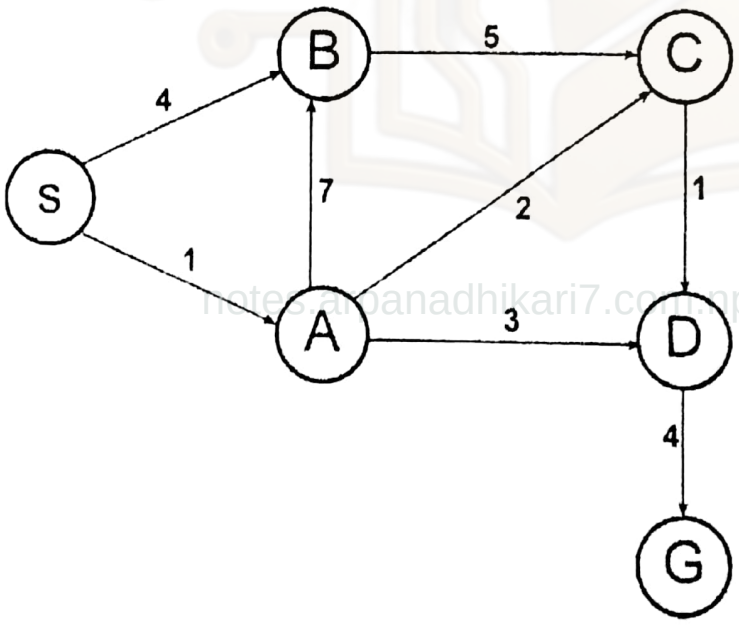
Attempt all the questions.

1. a) What is an intelligent agent? Write down the properties of agents. Explain goal based agents & utility based agents in brief. 8

   b) What is Constrain Satisfaction Problem (CSP). Solve the crypt-arithmetic problem as CSP. 7

$$\begin{array}{r} SEND \\ +\ MORE \\ \hline MONEY \end{array}$$

2. a) Using A* search algorithm, find the optimal route from S to G. 8

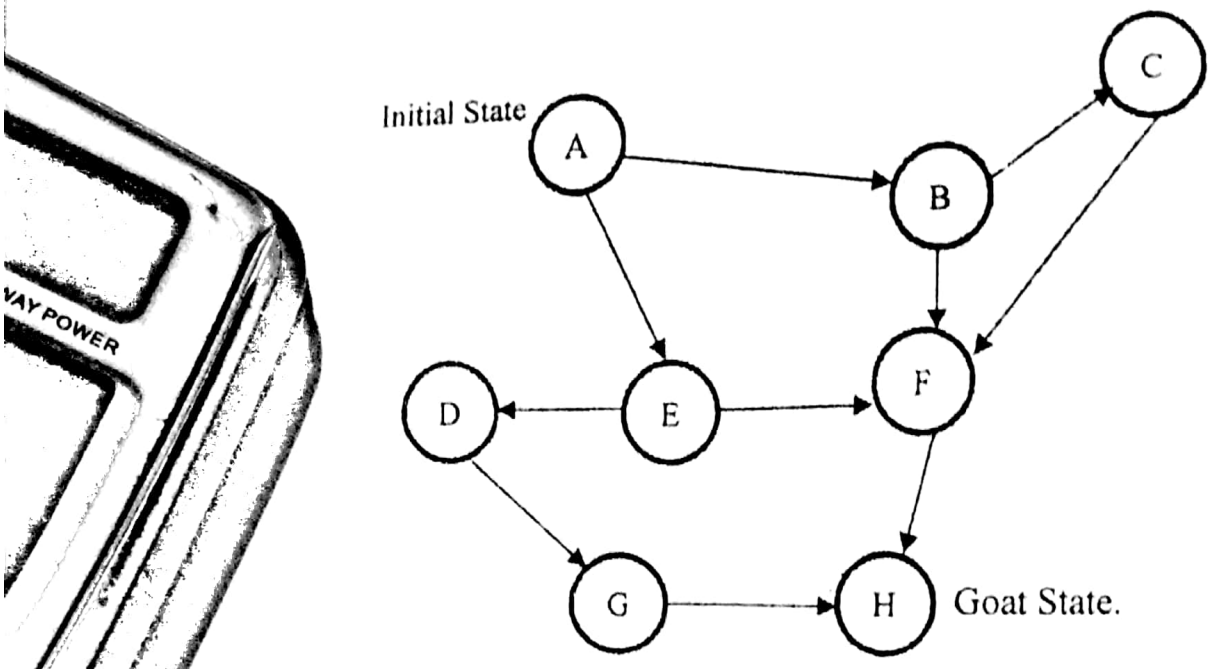


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   b) Consider a graph of cities as shown in below figure: 7

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i. Represent state space of this graph using prolog.
ii. Write a simple search algorithm in prolog to reach to the goal state from initial state.

3. a) What is knowledge representation? Define logical equivalence, tautology, contradiction and contingent with example.
   b) Represent the following statement using predicate logic.
      i. Anyone who buys carrots by the bushel owns either a rabbit or a grocery store.
      ii. Every dog chases some rabbit.
      iii. Someone who hates something owned by another person will not date that person.
      iv. If Mary does not own a grocery store, she will not date John.

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4. a) What is Deep Learning? Explain the types of stochastic neurons of Boltzmann machine?
   b) Why K-nearest Neighbors Algorithm is known as non-parametric lazy algorithm? Explain with example.

5. a) How can we find the right hyperplane in SVM? Using K-means clustering, cluster the following data into two clusters and show each step.
   {2, 4, 10, 12, 3, 20, 30, 11, 25}.
   b) What is a perceptron? How back propagation algorithm is used in learning and predicting data in machine learning?

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6. a) Draw the architecture of Expert System. How forward chaining and backward chaining assist the inference engine of Expert System? 7

b) What are the different types of ambiguities that accur in Natural Language Processing? Explain with examples 8

7. Write short notes on: (**Any two**) 2×5

a) Fuzzy learning

b) Best first search

c) Baye's Theorem

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